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Spatial Statistics Methods for the Analysis of Infectious Disease Clusters in Latin America

Abstract

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Spatial Statistics Methods for the Analysis of Infectious Disease *Clusters* in Latin America

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Abstract. The detection of infectious disease clusters through spatial statistics methods is a central strategy in modern epidemiological surveillance. However, no systematic synthesis exists that characterizes these methods within the Latin American context. This article presents a systematic review following the PRISMA 2020 guidelines, with searches conducted in Scopus and IEEE Xplore for the period 2015–2025. After an automated screening process and manual review, 76 articles were selected that explicitly employ spatial methods in infectious disease studies conducted in Latin America. The results show that the global Moran’s I index and LISA analysis are the most frequent methods (31.6% each), followed by SaTScan (23.7%) and Getis-Ord Gi* (13.2%). Brazil accounts for 88.2% of publications. The most studied diseases are dengue and arboviruses (23.7%), leishmaniasis (19.7%), tuberculosis (15.8%), and COVID-19 (14.5%). A methodological gap is identified in the use of hierarchical Bayesian spatio-temporal models, which are underrepresented in the region despite their well-recognized robustness. These findings guide the geospatial research agenda in public health for Latin America.

Keywords: spatial statistics · infectious diseases · Latin America · cluster detection · systematic review · PRISMA 2020 · geospatial epidemiology

1 Introduction

In the contemporary epidemiological context, infectious diseases continue to represent one of the greatest threats to public health and global socioeconomic stability. Their management demands not only clinical responses, but also the development of advanced analytical tools that allow their spread to be anticipated, modeled, and mitigated [20]. In recent decades, the integration of geospatial analysis into epidemiology has transformed the understanding of transmission dynamics, making it possible to observe how environmental and social factors interact within a given territory [39].

Spatial statistics has established itself as a fundamental pillar for identifying non-random patterns in the distribution of diseases. Its applications span four major areas: visualization, detection of global clustering, identification of *hotspots*, and risk factor modeling with adjustment for spatial heterogeneity [39]. Advanced methodologies such as Bayesian spatio-temporal models have demonstrated high effectiveness in the early detection of transmission clusters, as evidenced during the COVID-19 health crisis in 2020 [86,14]. These approaches provide health systems with an empirical basis for the optimal allocation of resources and the design of targeted intervention policies [36].

Latin America faces particular challenges due to its geographic heterogeneity, socioeconomic gaps, and the persistence of emerging and re-emerging diseases. Brazil, for example, recorded in 2024 its largest historical dengue epidemic with more than six million cases [62]. Colombia has accumulated more than 319,567 leishmaniasis cases over four decades, with hotspots concentrated in Amazonian and Pacific regions [3]. In Peru, geospatial studies in Lima have demonstrated that routine surveillance data are sufficient to identify high-incidence tuberculosis areas at the neighborhood scale [10].

Despite these advances, no systematic synthesis yet exists that comprehensively compares the spatial methodological approaches used in the Latin American context over the last decade. This gap makes it difficult to identify the most robust methods, the most studied diseases, and the recurrent limitations in the literature [20,30].

With the aim of providing a clear and structured overview, this article presents a systematic review of the scientific literature produced between 2015 and 2025. The main spatial statistics methods used for the analysis of infectious disease *clusters* in the Latin American context are examined and compared, discussing their scope, limitations, and relevance in modern epidemiological surveillance.

2 Methodology

2.1 Study Design

A systematic literature review was conducted following the guidelines established by PRISMA 2020 (*Preferred Reporting Items for Systematic Reviews and Meta-Analyses*) [49]. The review aimed to identify and analyze the spatial statistics methods used in studies on infectious diseases conducted in Latin America during the period 2015–2025.

2.2 Research Question

The research question was formulated using the PECO framework:

- **P (Population):** populations in Latin America and the Caribbean affected by infectious diseases.
- **E (Exposure):** application of spatial statistics methods in epidemiological analysis.
- **C (Comparison):** comparison among different spatial methods used for the detection and analysis of epidemiological patterns.
- **O (Outcome):** identification of clusters, spatial patterns, and factors associated with the geographic distribution of infectious diseases.

From this framework the following question was formulated: *What are the main spatial statistics methods used for the analysis of infectious disease clusters in Latin America, and what are their most frequent applications during the period 2015–2025?*

2.3 Information Sources

The bibliographic search was conducted in the Scopus and IEEE Xplore databases, due to their broad coverage of indexed scientific publications in the areas of public health, epidemiology, statistics, informatics, and geospatial analysis. Retrieved records were exported in RIS format, including title, authors, abstract, keywords, year of publication, journal, and source.

2.4 Search Strategy

The search was carried out using terms related to spatial statistics, epidemiology, and infectious diseases. The general search equation used was:

```
("spatial statistics" OR "spatial analysis" OR
"geostatistics" OR "hotspot analysis" OR
"SaTScan" OR "Moran index" OR "LISA" OR
"Kriging" OR "GWR" OR "DBSCAN")
AND
("infectious disease" OR "epidemiology" OR
"dengue" OR "malaria" OR "tuberculosis" OR
"COVID-19" OR "leishmaniasis")
AND
("Latin America" OR "South America" OR
"Peru" OR "Brazil" OR "Colombia" OR
"Mexico" OR "Bolivia" OR "Ecuador")
AND PUBYEAR > 2014
```

The search was limited to publications in English and Spanish published between 2015 and 2025.

2.5 Eligibility Criteria

Inclusion Criteria

- Scientific articles published between 2015 and 2025.
- Studies that explicitly employ spatial statistics methods.
- Research related to infectious diseases.
- Studies conducted in Latin American and Caribbean countries.
- Publications indexed in Scopus or IEEE Xplore.

Exclusion Criteria

- Studies conducted outside the Latin American context.
- Research without application of spatial methods.
- Review articles, editorials, letters to the editor, or conference abstracts.
- Duplicate records.
- Studies with insufficient information to determine eligibility.

2.6 Data Retrieval and Preparation

The bibliographic records obtained from Scopus and IEEE Xplore were exported in RIS format and integrated into a single database. Subsequently, a metadata normalization process was carried out to unify field names, correct inconsistencies, and eliminate duplicate records through comparison of normalized titles.

2.7 Tools Used

Various support tools were used for the development of the review. Mendeley was used for the management and verification of bibliographic references. RStudio was used for reading RIS files, integrating records, removing duplicates, applying automatic selection criteria, and generating preliminary descriptive statistics. The libraries `revtools`, `dplyr`, `stringr`, `purrr`, `ggplot2`, and `writexl` were used for automated information processing.

2.8 Automated Screening Process

Due to the large number of retrieved records, an automated procedure was developed in RStudio to carry out a first screening phase. The algorithm evaluated the content of titles, abstracts, and keywords using specialized dictionaries built from three categories: (1) spatial statistics methods, (2) infectious diseases, and (3) Latin American countries and regions.

Each record received a relevance score based on compliance with these criteria and was classified into one of three categories:

- **INCLUDE:** studies that simultaneously meet the thematic, methodological, and geographic criteria.
- **UNCERTAIN:** potentially relevant studies requiring additional manual review.
- **EXCLUDE:** studies that do not meet the established criteria.

2.9 Selection Process

The selection of studies was carried out in two phases. The first phase corresponded to the automated screening described above. The second phase consisted of the manual review of titles, abstracts, and full texts of studies classified as potentially eligible, in order to confirm their definitive inclusion in the systematic review. After this stage, the sample was composed of **76 articles**.

2.10 Methodological Quality Assessment

The methodological quality of the selected studies was assessed using the CASP (*Critical Appraisal Skills Programme*) tool, which classifies studies according to their quality level and risk of bias.

2.11 PRISMA 2020 Flowchart

The complete process of identification, screening, eligibility assessment, and inclusion is summarized in Figure 1.

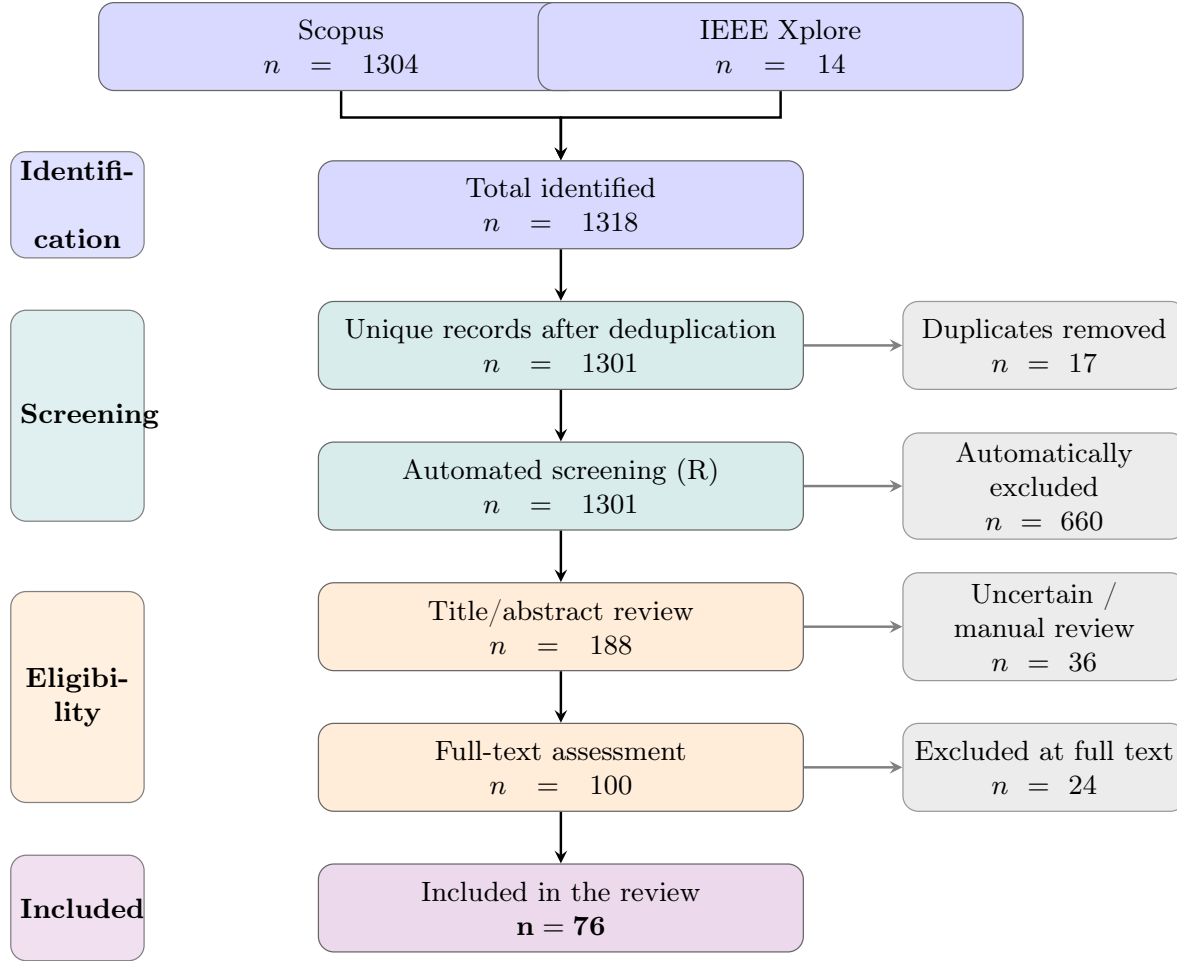


Fig. 1. PRISMA 2020 flow diagram — final sample $n = 76$.

3 Results

3.1 Selection Process Results

The search strategy applied in Scopus and IEEE Xplore initially identified 1,318 bibliographic records. Of these, 1,304 came from Scopus and 14 from IEEE Xplore. After removing 17 duplicate records through comparison of normalized titles, a consolidated database of 1,301 unique articles was obtained.

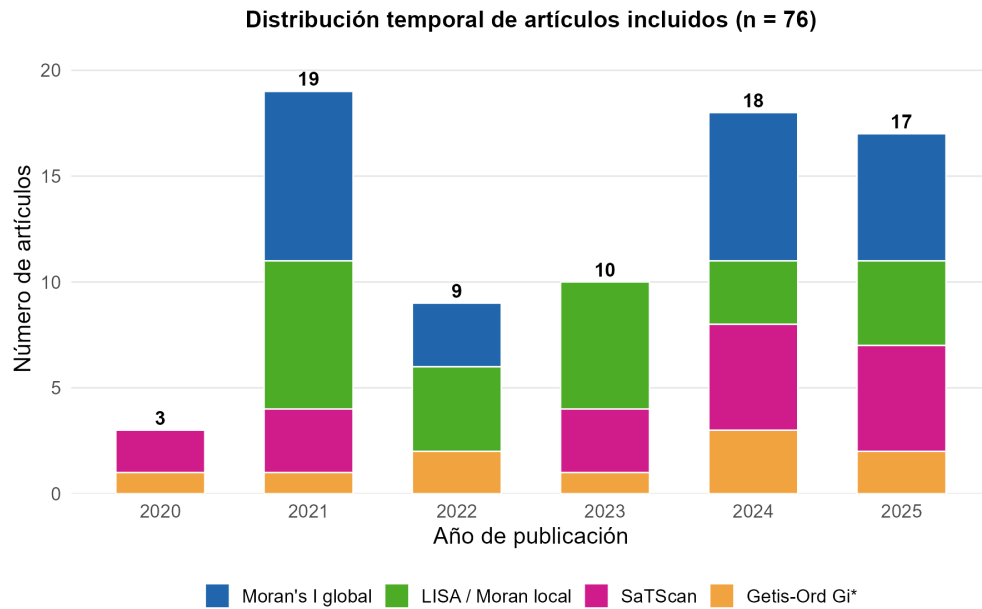
The automated screening process automatically discarded 660 articles and flagged 36 as uncertain for manual review. The 605 preliminarily included articles were subjected to successive prioritization stages that reduced the set to 188 articles. Of these, 100 high-relevance studies were selected for the full-text assessment phase. After detailed reading, 24 articles that did not fully meet the inclusion criteria were excluded, yielding a final sample of **76 articles**. The complete list of included studies is presented in Appendix A.

3.2 Temporal Distribution

The 76 included articles were published between 2020 and 2025 (Figure 2). A sustained increase in scientific output in the area is observed, with a marked rise starting in 2021. The years 2021, 2024, and 2025 account for the largest share of publications, with 19, 18, and 17 articles respectively, reflecting the surge in geospatial epidemiological studies in Latin America during and after the COVID-19 pandemic.

Table 1. Summary of the PRISMA 2020 selection process

Stage	Count
Records identified in Scopus	1,304
Records identified in IEEE Xplore	14
Total identified	1,318
Duplicates removed	17
Unique records	1,301
Automatically excluded	660
Uncertain for manual review	36
Preliminarily included	605
Prioritized articles	188
Selected for full-text review	100
Excluded at full text	24
Final included sample	76

**Fig. 2.** Temporal distribution of included articles ($n = 76$).

3.3 Distribution by Country

Brazil widely dominates the regional scientific output in this field, accounting for 67 of the 76 included articles (88.2%), followed at a distance by Mexico (3; 3.9%), Ecuador (2; 2.6%), and Colombia (2; 2.6%) (Figure 3). This concentration reflects both the magnitude of the infectious disease burden in Brazil and the established tradition of its academic institutions and epidemiological surveillance systems in the use of geospatial tools.

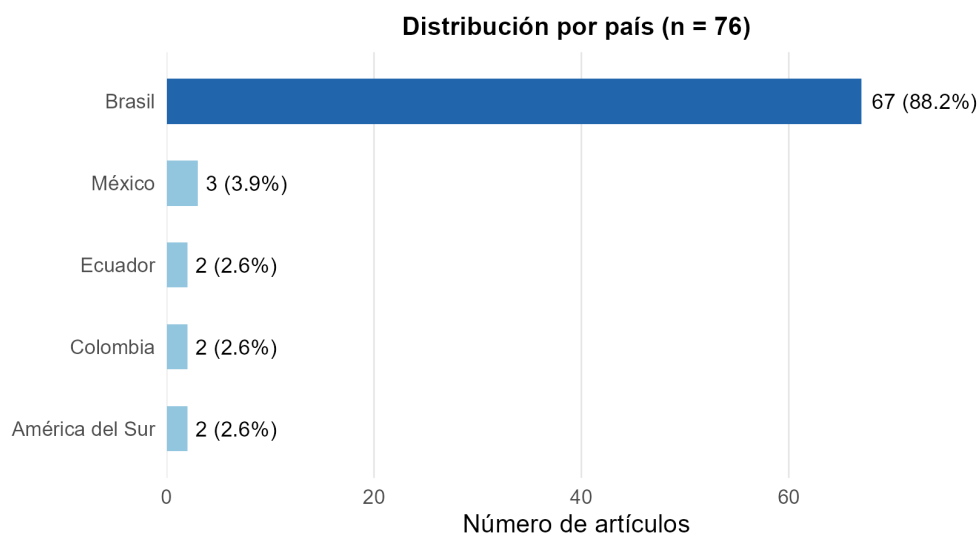


Fig. 3. Distribution by country ($n = 76$).

3.4 Distribution by Disease

The infectious diseases most studied through spatial statistics methods were dengue and arboviruses combined (18; 23.7%), leishmaniasis (15; 19.7%), tuberculosis (12; 15.8%), and COVID-19 (11; 14.5%) (Figure 4). A smaller proportion of studies was identified on leprosy (5), malaria (4), Chagas disease (2), schistosomiasis (2), and other infectious diseases.

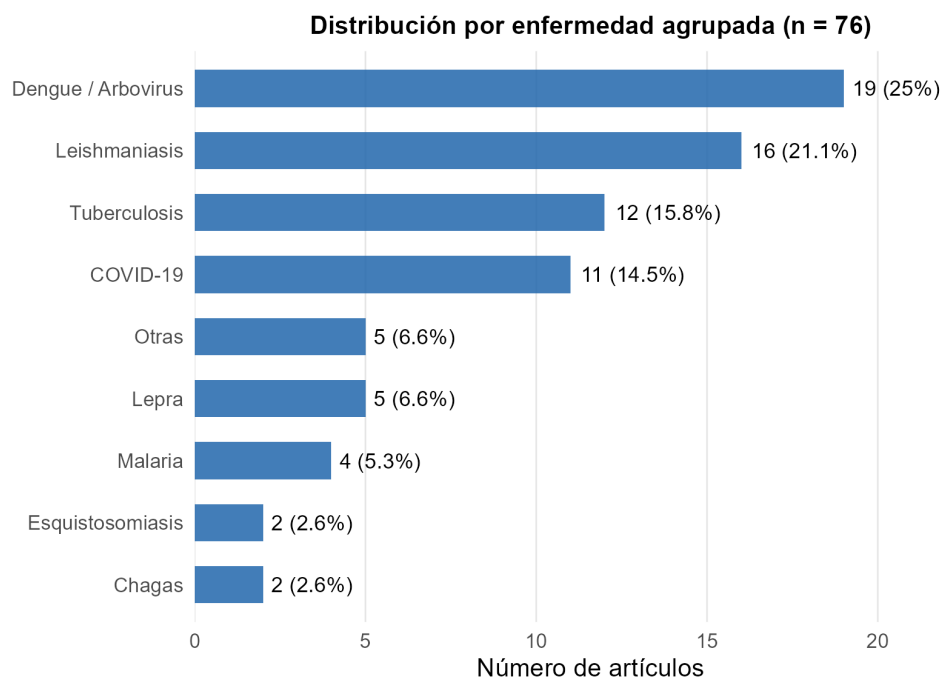


Fig. 4. Distribution by grouped disease ($n = 76$).

3.5 Spatial Methods Identified

The four spatial statistics methods identified in the final sample show a relatively balanced distribution between global and local autocorrelation techniques (Figure 5). The global Moran's I index and LISA (*Local Indicators of Spatial Association*) analysis are the most frequent, with 24 articles each (31.6% respectively), followed by SaTScan with 18 articles (23.7%) and Getis-Ord Gi* with 10 articles (13.2%).

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The identified studies show that Moran's I, LISA, SaTScan, and Getis-Ord Gi* are the predominant spatial methods used for the analysis of infectious diseases in Latin America. Their main applications and representative studies are summarized in Table 2.

Table 2. Spatial methods identified in the reviewed studies

Method	Main application and representative studies
Moran's I	Detection of global spatial autocorrelation and non-random geographical patterns [23,76,33,38,59].
LISA	Identification of local clusters and local spatial dependence [37,32,41,8,54].
SaTScan	Detection of statistically significant spatial and spatio-temporal clusters [79,68,80,51,16].
Getis-Ord Gi*	Identification of hotspots and coldspots of disease incidence [65,81,60,55,18].

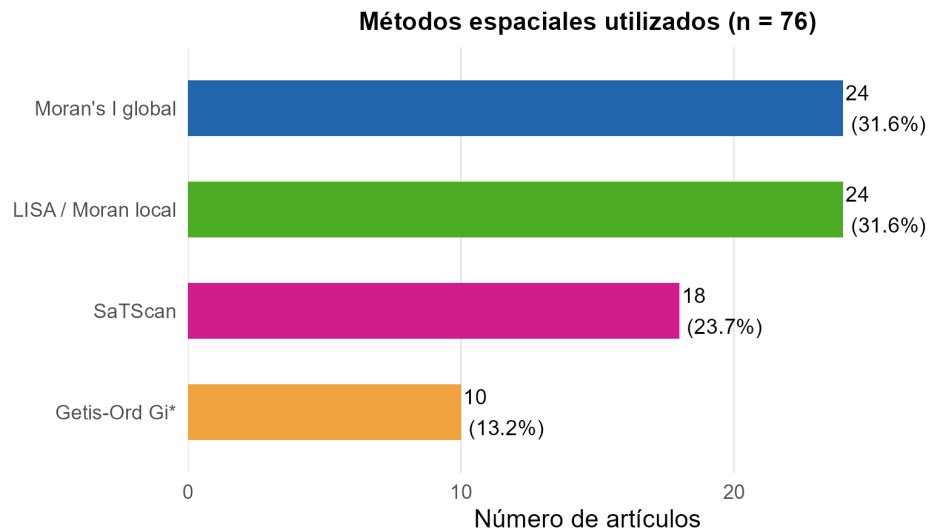


Fig. 5. Spatial methods used ($n = 76$).

3.7 Relationship Between Methods and Diseases

Figure 6 presents the heat map of the frequency of method–disease combinations. Leishmaniasis is studied primarily through global Moran's I and LISA, while tuberculosis shows the most diversified use of all four

methods. SaTScan concentrates its application on dengue and tuberculosis. Getis-Ord Gi* appears more frequently in studies of tuberculosis and leishmaniasis.

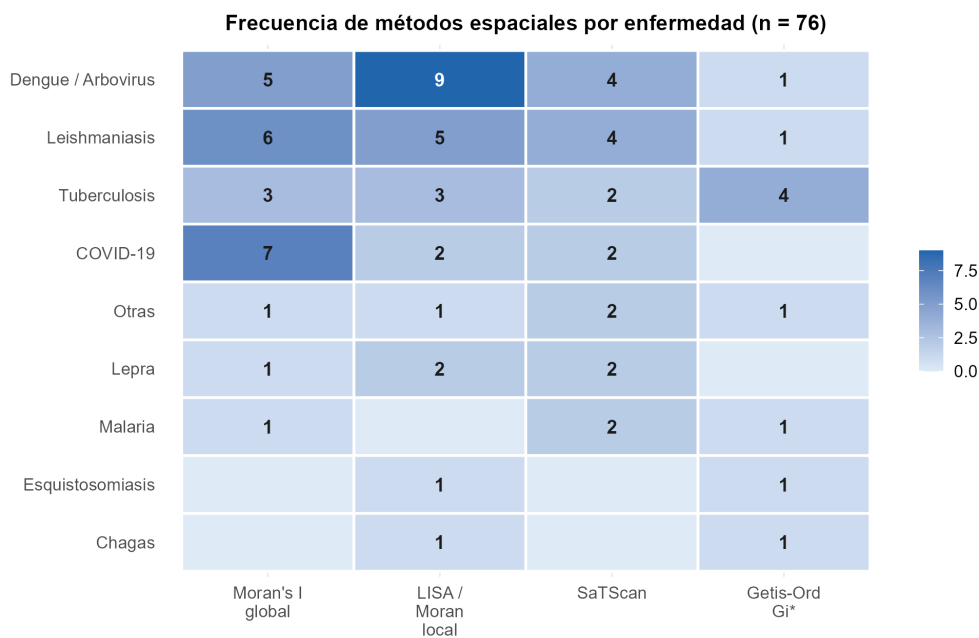


Fig. 6. Frequency of spatial methods by disease ($n = 76$).

4 Discussion

The results of this systematic review confirm that spatial statistics has established itself as an indispensable tool for the surveillance and analysis of infectious diseases in Latin America during the period 2015–2025. The concentration of studies in Brazil, the predominance of dengue and leishmaniasis, and the preference for global and local autocorrelation methods reflect both the regional epidemiological burden and the institutional capacities of each country to produce geospatial research.

4.1 Brazil's Dominance in Regional Scientific Output

The overwhelming proportion of studies from Brazil (88.2%) responds to well-documented structural factors. The country has robust and publicly accessible epidemiological surveillance systems, such as DATASUS, which facilitate the retrieval of georeferenced data at the municipal scale [79]. Likewise, the high burden of neglected tropical diseases, the ecological diversity of the Amazonian territory, and the institutional resources of federal and state universities explain this concentration [76,37]. This finding is consistent with previous reviews noting that Latin American production in geospatial epidemiology originates primarily in Brazil, Colombia, and Mexico [20].

The scarce representation of countries such as Peru, Bolivia, Paraguay, and much of Central America constitutes an important limitation of the available literature, given that many of these territories harbor endemic foci of leishmaniasis, malaria, and tuberculosis without the analytical coverage that their public health needs would demand [30].

4.2 Spatial Methods: Trends and Applications

The spatial autocorrelation methods, global Moran's I and LISA, are the most used in the analyzed sample (31.6% each). This prevalence reflects their straightforward implementation in platforms such as GeoDa, R,

and QGIS, as well as their direct interpretability for public health decision-makers. The global Moran's I index provides a synthetic measure of the non-random spatial distribution of a disease, while its local complement—LISA—enables the identification of specific municipalities or regions that form significant clusters of high or low incidence [39].

SaTScan (23.7%) differs from the above by incorporating the temporal dimension, allowing the detection of clusters that emerge or intensify during specific periods. This feature makes it especially valuable for the analysis of diseases with marked seasonal dynamics, such as dengue and leptospirosis, or for evaluating the impact of public health interventions [79,5]. Getis-Ord G_i^* (13.2%) complements the others by identifying geographic *hotspots* of high local incidence with statistical significance, being particularly useful in studies at the municipal or neighborhood scale where internal heterogeneity is high [81].

It is worth noting that no study in the final sample exclusively employed hierarchical Bayesian spatio-temporal models, despite the international literature recognizing them as the most suitable for modeling uncertainty in small areas with low counts [86,14]. This methodological gap represents a development opportunity for Latin American epidemiological research, especially for the analysis of re-emerging diseases and new events of public health interest [46].

4.3 Most Studied Diseases

Dengue and associated arboviruses account for the largest proportion of studies (23.7%), consistent with the historic epidemic recorded in Brazil in 2024 and the endemic situation in the region [62]. The identified studies demonstrate that spatial methods allow not only mapping the distribution of cases, but also linking transmission patterns with socio-environmental variables such as vegetation cover, temperature, and drinking water coverage [74,21].

Leishmaniasis (19.7%) ranks second, reflecting its endemic nature in northern and northeastern Brazil, Colombia, and Peru [76,3]. The reviewed studies predominantly use Moran's I and LISA to identify clusters in Amazonian and Pacific regions with high vector exposure [32,78].

Tuberculosis (15.8%) appears as the disease with the greatest methodological diversity in its geospatial analysis: studies were identified using all four analyzed methods, including SaTScan for the detection of spatio-temporal clusters linked to the COVID-19 pandemic and the disruption of control programs [79,54]. This finding confirms that tuberculosis remains a sensitive indicator of territorial social inequities and a productive field for geospatial research.

4.4 Limitations of the Review

This review presents limitations that should be considered when interpreting its results. First, the search was restricted to two databases, which may have excluded relevant studies published in regional repositories or in Portuguese without indexing in Scopus. Second, the classification of spatial methods was based primarily on information declared in titles and abstracts, which may have underestimated the presence of complementary approaches described only in the methods sections of the articles. Finally, the geographic concentration of the sample in Brazil limits the generalizability of the findings to the Latin American region as a whole.

5 Conclusions

This systematic review demonstrates that spatial statistics methods have established themselves as indispensable tools for the epidemiological surveillance of infectious diseases in Latin America during the period 2015–2025. The analysis of the 76 included articles allows the following conclusions to be drawn.

First, the global Moran's I index and LISA analysis are the most widely used methods in the region (31.6% each), reflecting their accessibility in platforms such as GeoDa, R, and QGIS, as well as their direct interpretability for public health decision-makers. SaTScan stands out for incorporating the temporal dimension, making it the preferred tool for diseases with marked seasonal dynamics such as dengue, while Getis-Ord G_i^* is particularly useful for identifying local *hotspots* at the municipal scale.

Second, Brazil concentrates the largest regional scientific output (88.2%), supported by robust publicly accessible surveillance systems such as DATASUS and a high burden of tropical diseases. This concentration

contrasts with the scarce representation of countries such as Peru, Bolivia, and much of Central America, where endemic foci of leishmaniasis, malaria, and tuberculosis persist without geospatial analytical coverage proportional to their disease burden.

Third, dengue and arboviruses (23.7%), leishmaniasis (19.7%), and tuberculosis (15.8%) are the diseases most studied through these methods, consistent with the epidemiological burden of the region. The COVID-19 pandemic also accelerated the development and application of spatio-temporal models, demonstrating the value of these approaches for decision-making in public health emergencies.

Finally, a significant methodological gap is identified: hierarchical Bayesian spatio-temporal models, internationally recognized as the most appropriate for modeling uncertainty in small areas with low counts, are underrepresented in the Latin American literature. Bridging this gap, together with expanding the geographic coverage of studies toward countries with less tradition in geospatial research, constitutes the main pending agenda for spatial epidemiology in the region.

In summary, the systematic integration of spatial statistics into epidemiological surveillance programs in Latin America represents a key strategy for the early detection of outbreaks, the prioritization of resources, and the design of targeted interventions. Strengthening these analytical capacities, especially in countries with greater health vulnerability, is essential to prevent and mitigate the impact of future public health emergencies.

A Included Studies

The systematic review finally included 76 studies covering the period 2020–2025. These studies were used for the quantitative and qualitative synthesis presented in this review.

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Table 3. List of included studies.

N° Study Key	Citation	N° Study Key	Citation
1 FloresSacoto2025	[23]	39 Alves2021	[50]
2 Miranda2024	[76]	40 Guo2020	[29]
3 DeLima2025	[37]	41 Alves2022	[2]
4 Souza2023	[79]	42 DzulManzanilla2021	[21]
5 Konczykcki2025	[32]	43 Silva2023	[67]
6 Silva2024	[65]	44 Berra2020	[6]
7 Tavares2024	[81]	45 Siqueira2021	[74]
8 Lalangui2024	[33]	46 doCarmo2020	[11]
9 deLima2024	[38]	47 daSilva2022	[63]
10 Matos2025	[41]	48 Wangdi2022	[85]
11 Silva2025	[68]	49 Adegboye2021	[1]
12 Santos2024	[59]	50 Pereira2021	[52]
13 AlmeidaSouza2024	[80]	51 Pereira2021b	[53]
14 BirchenallJimenez2024	[8]	52 SoaresSantana2021	[58]
15 Perotto2025	[54]	53 doSocorroCarvalhoMiranda2022	[77]
16 DaudtLemos2025	[19]	54 GaytcanCamarillo2021	[26]
17 Paz2023	[51]	55 daSilva2023	[66]
18 Gomes2023	[27]	56 Schultes2021	[61]
19 dosSantos2024	[60]	57 Miranda2021	[45]
20 Saavedra2025	[16]	58 FcalixRocha2023	[56]
21 Lopes2025	[40]	59 Nina2023	[70]
22 Pungartnik2025	[55]	60 Costa2021	[69]
23 deFreitas2024	[25]	61 DeOliveira2021	[47]
24 MedeirosSilva2025	[42]	62 Huang2021	[31]
25 Gomes2023b	[28]	63 Skalinski2022	[75]
26 CharlesLeija2024	[13]	64 Braga2021	[48]
27 Veloso2025	[83]	65 Braz2021	[9]
28 Bermudi2025	[5]	66 Coutinho2022	[17]
29 Freitas2024	[24]	67 daSilvaSoeiro2022	[72]
30 Damasceno2025	[18]	68 Medeiros2023	[4]
31 deMelo2024	[44]	69 Echenique2021	[22]
32 daSilva2024	[64]	70 Leandro2022	[35]
33 Biazussi2025	[7]	71 CortesEscamilla2022	[15]
34 Silveira2024	[73]	72 deMelo2023	[43]
35 Rocha2025	[57]	73 Ueno2021	[82]
36 VieiraDuarte2024	[84]	74 Castro2021	[12]
37 Leandro2024	[34]	75 Queiroz2021	[71]
38 deSousa2024	[78]		

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